

A New Framework for Creating Model-Assisted Semiparametric Naïve Bayes (MASNB) Classifiers

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1 Introduction

Classification remains a central problem in statistical learning and artificial intelligence, with applications spanning biomedical diagnostics, financial risk assessment, and natural language processing. Despite the proliferation of powerful algorithms—ranging from ensemble methods [2] to deep neural networks—no single classifier dominates across all data regimes. This motivates the continued development of novel frameworks that expand the theoretical and practical boundaries of supervised learning.

Naïve Bayes (NB) classifiers occupy a unique niche due to their computational efficiency, scalability, and robustness to irrelevant features [10]. However, their performance is critically dependent on the quality of density estimation for class-conditional feature distributions. Existing NB variants—parametric, non-parametric, and semiparametric—share a fundamental limitation: they estimate the conditional densities $P(x_i | C_j)$ **independently** for each class j . This independence assumption, while computationally convenient, discards potentially valuable inter-class information and can lead to suboptimal estimates, particularly in high-dimensional settings or when class distributions are overlapping.

This project proposes a **novel Model-Assisted Semiparametric Naïve Bayes (MASNB) framework** that overcomes this limitation. The core innovation is to replace independent estimation with **joint estimation** of all class-conditional densities using empirical likelihood-based density ratio models [11, 16]. By explicitly modeling the inter-class relationships via density ratio functions, the MASNB framework achieves three fundamental advances:

1. **Theoretical:** It provides a unified semiparametric foundation for NB classifiers, bridging parametric and non-parametric paradigms through the lens of density ratio modeling.
2. **Methodological:** It introduces new estimators for $P(x_i | C_j)$ that leverage full-sample information, improving efficiency and robustness.
3. **Practical:** It yields a family of classifiers that are expected to outperform existing NB variants, especially in small-sample and high-dimensional settings.

2 Background and Problem Formulation

2.1 The Naïve Bayes Classifier

Let $Y \in \{C_1, \dots, C_K\}$ denote the class label and $\mathbf{x} = (x_1, \dots, x_m)^\top \in \mathbb{R}^m$ be a vector of continuous features. The NB classifier is defined by the posterior probability:

$$P(C_j | \mathbf{x}) \propto P(C_j) \prod_{i=1}^m f_j(x_i), \quad (1)$$

where $f_j(x_i) = P(x_i | C_j)$ is the class-conditional density of feature x_i . The “naïve” assumption of conditional independence enables the factorization, drastically reducing the number of parameters to be estimated.

2.2 Existing Approaches and Their Limitations

Existing NB classifiers differ in how they estimate $f_j(x_i)$:

- **Parametric** (e.g., Gaussian NB): Assumes $f_j(x_i) = \mathcal{N}(\mu_{ji}, \sigma_{ji}^2)$, with parameters estimated via maximum likelihood. This is efficient but brittle when distributional assumptions are violated.
- **Non-parametric** (e.g., Kernel NB): Estimates $f_j(x_i)$ via kernel density estimation (KDE):

$$\hat{f}_j(x) = \frac{1}{n_j h} \sum_{s: y_s = C_j} K\left(\frac{x - x_{is}}{h}\right), \quad (2)$$

where n_j is the class-specific sample size and h is the bandwidth [12, 20]. This is flexible but data-inefficient, as it uses only class-specific observations and ignores cross-class information.

- **Semiparametric**: Typically refers to a hybrid approach that uses KDE for continuous features and parametric models for discrete features. This does not address the fundamental issue of independent class-wise estimation for continuous features.

The critical gap is that all existing estimators treat each class’s density as an independent entity, even when the classes share underlying structure. In many real-world problems, densities across classes are related; for example, they may differ by a location shift or a scale change. Ignoring these relationships discards valuable information and reduces estimation efficiency.

3 The Model-Assisted Semiparametric Naïve Bayes (MASNB) Framework

3.1 Theoretical Foundation: Density Ratio Models

Without loss of generality, consider a single continuous feature X and let C_1 be a designated baseline class. Define the **density ratio function**:

$$R_j(x; \theta_j) = \frac{f_j(x)}{f_1(x)}, \quad j = 2, \dots, K, \quad (3)$$

where $f_j(x) = P(x | C_j)$ and θ_j is a finite-dimensional parameter. The ratio must satisfy the integral constraint:

$$\int_{\Omega} R_j(x; \theta_j) f_1(x) dx = 1, \quad (4)$$

ensuring that $f_j(x)$ integrates to unity.

The existence of such ratio functions is well-established in the semiparametric literature. For instance, logistic regression arises from the canonical choice:

$$R_j(x; \boldsymbol{\theta}_j) = \exp(\alpha_j + \beta_j x), \quad (5)$$

where $\boldsymbol{\theta}_j = (\alpha_j, \beta_j)^\top$. This is a special case of the **two-sample density ratio model** introduced by [16].

3.2 The MASN B Reformulation

Substituting $f_j(x_i) = R_j(x_i; \boldsymbol{\theta}) f_1(x_i)$ into the NB classifier yields:

$$P(C_j | \mathbf{x}) \propto P(C_j) \prod_{i=1}^m R_j(x_i; \boldsymbol{\theta}) f_1(x_i). \quad (6)$$

This reformulation has profound implications:

1. **Joint Estimation:** Instead of estimating each f_j independently, we now estimate the **baseline density** f_1 and the **ratio functions** R_j jointly using all available data. This leverages inter-class relationships to improve estimation accuracy.
2. **Dimension Reduction:** The parameter vector $\boldsymbol{\theta}$ is typically low-dimensional (e.g., $2(K-1)$ for logistic ratios), which imposes a structured relationship between densities and reduces the effective number of parameters.
3. **Semiparametric Efficiency:** Under mild regularity conditions, the empirical likelihood-based estimators for f_1 and $\boldsymbol{\theta}$ achieve the semiparametric efficiency bound [16], meaning they are asymptotically optimal.

3.3 Estimation Methods

The MASN B framework accommodates two broad classes of estimators:

3.3.1 Empirical Likelihood-Based Estimators (EL-MASN B)

The empirical likelihood (EL) approach [11] maximizes the multinomial likelihood subject to the density ratio constraints. For the two-sample case ($K = 2$), the EL estimator for f_1 and $\boldsymbol{\theta}$ is obtained by maximizing:

$$\mathcal{L}_{\text{EL}}(\boldsymbol{\theta}, p_1, \dots, p_n) = \prod_{s=1}^n p_s \quad (7)$$

subject to:

$$p_s \geq 0, \quad \sum_{s=1}^n p_s = 1, \quad \sum_{s=1}^n p_s [R(x_s; \boldsymbol{\theta}) - 1] = 0, \quad (8)$$

where p_s are probability masses assigned to each observation. This yields the baseline density estimator:

$$\hat{f}_1(x) = \sum_{s=1}^n p_s \delta_{x_s}(x), \quad (9)$$

and the density ratio estimator $\hat{R}(x; \hat{\theta})$. For $K > 2$, the multi-class extension follows [21] and [14], involving multiple ratio constraints.

3.3.2 Semiparametric Kernel-Based Estimators (SK-MASNB)

The semiparametric kernel estimator [4, 18] combines the parametric density ratio model with non-parametric KDE. The baseline density is estimated as:

$$\hat{f}_1(x) = \frac{1}{nh} \sum_{s=1}^n \frac{K\left(\frac{x-x_s}{h}\right)}{\sum_{j=1}^K \pi_j R_j(x_s; \hat{\theta})}, \quad (10)$$

where $\pi_j = P(C_j)$ and $\hat{\theta}$ is obtained by maximizing a semiparametric likelihood. The class-conditional densities are then:

$$\hat{f}_j(x) = \hat{R}_j(x; \hat{\theta}) \hat{f}_1(x). \quad (11)$$

This estimator is more flexible than the EL estimator because it uses kernel smoothing, but it retains the efficiency gains from the density ratio model.

3.4 Specific Contributions and Subfamilies

The MASNB framework defines a **family of classifiers**, each characterized by:

1. A chosen ratio function $R_j(x; \theta)$;
2. An estimation method (EL, SK, or other);
3. (For ordinal data) a mechanism for structuring class relationships (e.g., proportional odds).

Initial Development (Phase I): We will focus on the exponential ratio function $R_j(x; \theta_j) = \exp(\alpha_j + \beta_j x)$ and develop two subfamilies:

- **Binary MASNB:** Using the two-sample EL estimator [16, 13] and the two-sample SK estimator [4, 18].
- **Multi-class MASNB:** Using the multi-sample EL estimator [21, 14] and a novel multi-sample SK estimator, which we will derive by extending the work of [4] and [18] to $K > 2$.

Future Development (Phase II): We will investigate alternative ratio functions, including:

- Probit link: $R_j(x; \theta) = \Phi(\alpha_j + \beta_j x)$;
- Complementary log-log: $R_j(x; \theta) = 1 - \exp(-\exp(\alpha_j + \beta_j x))$;
- Log-linear models for ordinal data: $R_j(x; \theta) = \exp(\alpha_j + \beta_j x)$ with constraints $\alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_K$.

Each choice yields a distinct classifier with unique theoretical properties and empirical performance characteristics.

Criterion	Parametric NB	Kernel NB	MASNB (Proposed)
Estimation paradigm	Independent	Independent	Joint
Inter-class relationships	Ignored	Ignored	Explicitly modeled
Sample efficiency	High	Low	High (due to borrowing strength)
Flexibility	Low	High	High (semiparametric)
Theoretical guarantees	Standard MLE	Asymptotic consistency	Semiparametric efficiency
Parameter count	$O(Km)$	$O(1)$ (bandwidth)	$O(Km)$ (structured)

4 Comparison to Existing Work

The MASNB framework is **fundamentally different** from existing NB variants in the following respects:

The key innovation is the **joint estimation** paradigm, which sets MASNB apart from all existing NB classifiers. This is not a minor modification but a fundamental reconceptualization of how class-conditional densities should be estimated.

5 Expected Contributions and Broader Impact

5.1 Theoretical Contributions

1. Formalization of a unified semiparametric framework for NB classification that encompasses and generalizes existing parametric and non-parametric approaches.
2. Derivation of the asymptotic properties (consistency, efficiency, and convergence rates) of the proposed EL and SK estimators within the NB context.
3. Characterization of the conditions under which density ratio models provide substantial efficiency gains over independent estimation.

5.2 Methodological Contributions

1. A suite of new classifiers (binary and multi-class, EL and SK) that are ready for implementation and benchmarking.
2. Practical algorithms for parameter estimation, including computational strategies for handling high-dimensional features.
3. Extensions to ordinal classification and mixed data types (continuous and discrete).

5.3 Practical Impact

MASNB classifiers are expected to outperform existing NB variants, especially when:

- Class distributions are overlapping;
- Sample sizes are small relative to the number of features;
- Inter-class relationships are strong and structured.

This has direct implications for applications in genomics, text classification, medical diagnostics, and any domain where efficient and robust classification is critical.

6 Conclusion

The Model-Assisted Semiparametric Naïve Bayes (MASNB) framework represents a significant departure from the conventional independent estimation paradigm that has dominated NB research for decades. By leveraging empirical likelihood-based density ratio models, MASNB enables the joint estimation of class-conditional densities, explicitly capturing inter-class relationships and improving estimation efficiency.

This project will deliver both theoretical insights and practical tools, advancing the state of the art in probabilistic classification. The initial focus on exponential ratio functions and two estimation paradigms (EL and SK) provides a solid foundation for a broader research program exploring alternative ratio functions, ordinal extensions, and mixed data types. MASNB is not merely a new classifier—it is a new lens for understanding and solving the classification problem.

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